Evidence against per-feature thresholds in LLMs

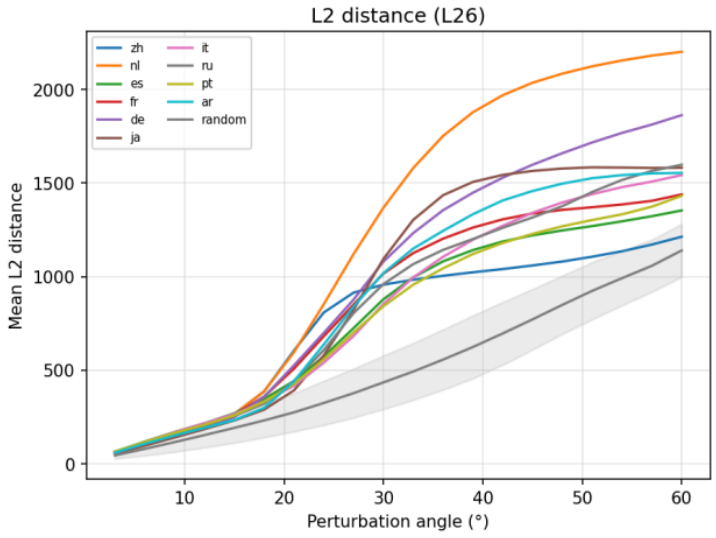
We think our results are evidence against superposition with error correction. We first present results in favour of error correction generally, and then show that this error correction does not support features in superposition.

Under the linear representation hypothesis (LRH), large language model (LLM) features are vectors and the model is particularly sensitive to these vectors. Under superposition, there are more feature vectors than there are dimensions in the residual stream. It is likely that (deep) computation in superposition requires error correction (Mathematical Models of Computation in Superposition).

A simple way LLMs could implement error correction is via per-feature thresholds: each feature direction d_i has an associated threshold t_i. The component of an activation along direction d_i only propagates downstream if it exceeds t_i. Under this picture, directions that aren't aligned with any feature (including random directions) have larger effective thresholds than feature directions. Perturbing an activation along a non-feature direction, a downstream response appears only once the perturbation's projection onto some feature crosses that feature's threshold.

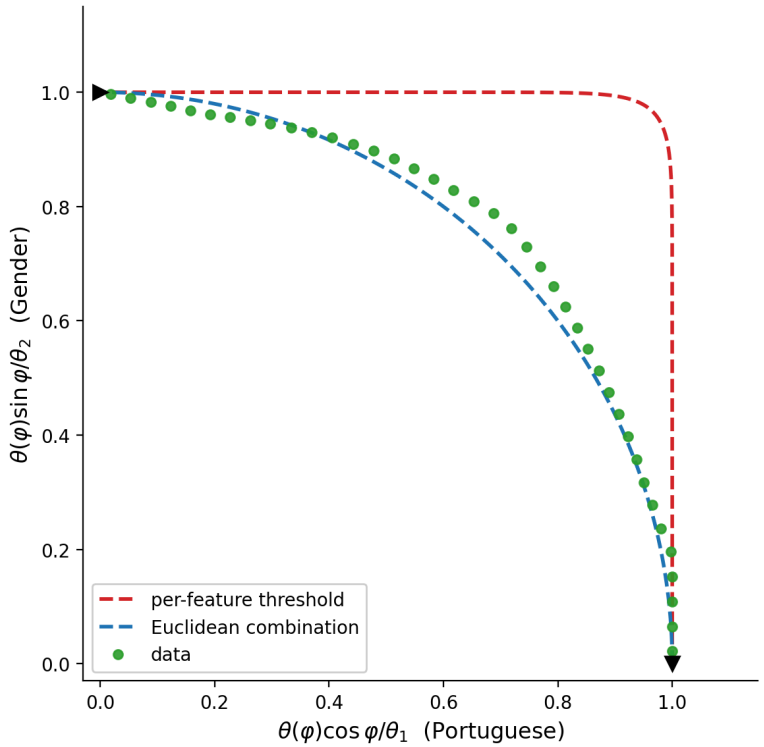
By perturbing the activations of LLMs and measuring the downstream effect, we observe results that appear consistent with this picture. Concretely, we extract residual stream activations at an early layer, perturb them while holding the norm constant, propagate the perturbed activation forward, and measure the distance between perturbed and unperturbed activations downstream.

We do this procedure for (i) contrastive directions, which we take to be an approximation of feature directions, and (ii) random directions:



There is a region of suppressed response for both (activation plateau), and this region is longer for random directions. In other words, a stronger perturbation is required to break out of the plateau when perturbing along a random direction. This effect is robust to the choice of layer of perturbation, layer of measurement, norm-matched and additive perturbation, and choice of downstream response measure (we test L^2 and cosine distance in activation space, and KL-divergence in the logits).

We will now argue that the empirical results we see are not consistent with error correction supporting features in superposition. Per-feature threshold error correction would predict that perturbing along combinations of two directions produces a downstream response only when the perturbation's projection onto either of the directions was enough to surpass its respective threshold. We can test this empirically. Consider two (contrastive) directions d_1 and d_2. First, we perform norm-matched perturbations towards each of them, and measure the angles at which we get a downstream response, Θ_1 and Θ_2. Then, we perform a norm-matched perturbation towards $\cos(\phi)d_1 + sin(\phi)d_2$, sweep $\phi \in [0, \pi/2]$, and measure the angle at which we get a downstream response, $\theta(\phi)$. We plot each measured breaking perturbation as its components along $d_1$and d_2, normalized by the respective threshold:



This example is for contrastive feature directions for Portuguese and gender on Gemma-9B (note that steering along the combination of these two directions works as expected, producing Portuguese-language feminine completions from English-language masculine inputs), measuring downstream response as L^2distance; we've found this holds for other contrastive pairs across different LLMs and measuring cosine distance and KL-divergence.

The error-correction-like behavior we observe appears to be a threshold on the Euclidean norm of the perturbation, which suggests that the model does not privilege the individual feature directions. To formalize this, we model the downstream model response R to perturbation v as

$$R(v) = F\left(\sum_i w_i |d_i^T v|^n\right)$$

where d_i are unit directions with weights w_i, and F is a convex function.

For perturbations along two directions, this functional form predicts that the perturbation angle for which there is a downstream response satisfies

$$\left(\frac{\theta(\phi)\cos\phi}{\theta_1}\right)^n + \left(\frac{\theta(\phi)\sin\phi}{\theta_2}\right)^n = 1,$$

a superellipse of exponent n in normalized coordinates. We fit this single parameter by minimizing the squared residual of this equation over the measured $(\theta(\phi)\cos\phi,\ \theta(\phi)\sin\phi)$ points. The fitted n is consistently close to 2 across conditions: different contrastive pairs; perturbations along combinations of random (orthogonalized) directions in higher dimensions, not just pairs; different models; different perturbation and measurement layers; and all three response metrics. Median fitted n sits between about 2.0 and 2.2, with per-pair fit residuals of order 1%.

If we set $n = 2$ the response function reduces to

$$R(v) = F\left(\sum_i w_i \,(d_i^\top v)^2\right) = F\left(v^\top A v\right), \qquad A = \sum_i w_i \, d_i d_i^\top,$$

a convex function of a positive semidefinite quadratic form.

Put differently, the response depends on v only through a smooth quadratic form. Its iso-response surfaces are ellipsoids, with a number of principal axes at most the residual stream's dimension. This means that at most that number of features can be privileged directions of the response function. This is in tension with LRH + superposition + error correction.